

OCI Frequently Asked Questions (FAQ)

1. What is OCI?

Oracle Cloud Infrastructure (OCI) is Oracle's cloud platform that provides Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) offerings for running enterprise workloads.

2. What are the main OCI service categories?

- Compute
 - Storage
 - Networking
 - Database
 - Security & Identity
 - Monitoring & Management
 - Containers & Kubernetes (OKE)
 - Analytics & AI
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3. What is a Tenancy in OCI?

A tenancy is a secure and isolated partition within OCI where all cloud resources are created and managed.

Example:

A company creates one OCI tenancy and manages all departments, projects, users, and cloud resources within it.

4. What are Compartments?

Compartments are logical containers used to organize and isolate OCI resources.

Example:

- Production Compartment
- Development Compartment
- Testing Compartment

Benefits:

- Resource organization
- Access control
- Cost tracking

5. What is a Region?

A Region is a localized geographic area containing one or more Availability Domains.

Example:

- Ashburn, Virginia
 - Phoenix, Arizona
 - Mumbai, India
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6. What is an Availability Domain (AD)?

An Availability Domain is an isolated data center within a region.

Benefits:

- Fault tolerance
 - High availability
 - Disaster recovery
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7. What is a Fault Domain?

A Fault Domain is a logical grouping of hardware within an Availability Domain.

Benefits:

- Protects against hardware failures
 - Improves application availability
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8. What is a Virtual Cloud Network (VCN)?

VCN is OCI's virtual network.

Similar to a traditional data center network.

Components include:

- CIDR Block
 - Subnets
 - Route Tables
 - Security Lists
 - Gateways
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9. Difference between Public and Private Subnet?

Public Subnet

Has Internet access

Uses Internet Gateway

Web servers usually placed here

Private Subnet

No direct Internet access

Uses NAT Gateway

Databases usually placed here

10. What is an Internet Gateway?

Allows communication between VCN resources and the public Internet.

Example: Public web application access.

11. What is a NAT Gateway?

Allows private instances to access the Internet without exposing them publicly.

Example: Downloading OS patches.

12. What is a Service Gateway?

Provides private access to Oracle services without traversing the Internet.

Example: Accessing Object Storage privately.

13. What is Dynamic Routing Gateway (DRG)?

DRG connects OCI networks with:

- On-Premises Data Centers
- Other OCI Regions
- Third-party Clouds

Used for hybrid cloud connectivity.

14. What is FastConnect?

Dedicated private connection between OCI and on-premises environment.

Benefits:

- Low latency
- High bandwidth

- Secure communication
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15. What is a Compute Instance?

A virtual machine running applications in OCI.

Types:

- VM
 - Bare Metal
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16. Difference between VM and Bare Metal?

VM	Bare Metal
Shared physical host	Dedicated server
Virtualized	No hypervisor
Lower cost	High performance

17. What is Block Volume?

Persistent storage attached to compute instances.

Similar to a hard disk.

Features:

- Backup
 - Clone
 - Resize
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18. What is Object Storage?

Highly scalable storage for unstructured data.

Examples:

- Backups
 - Images
 - Videos
 - Logs
-

19. Difference between Block, Object and File Storage?

Block Storage	Object Storage	File Storage
VM disks	Backup repository	Shared file system
High IOPS	Unlimited scalability	NFS access

20. What is OCI Load Balancer?

Distributes incoming traffic across multiple backend servers.

Benefits:

- High Availability
 - Scalability
 - Fault Tolerance
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21. Difference between Public and Private Load Balancer?

Public LB	Private LB
Internet-facing	Internal traffic only
Public IP	Private IP

22. What are Health Checks?

Load Balancer continuously checks backend server health.

Unhealthy servers are automatically removed from traffic routing.

23. What is SSL Termination?

Load Balancer decrypts HTTPS traffic before forwarding requests to backend servers.

Benefits:

- Reduced backend workload
 - Centralized certificate management
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24. What is Identity and Access Management (IAM)?

IAM controls authentication and authorization.

Components:

- Users
 - Groups
 - Policies
 - Dynamic Groups
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25. What is a Dynamic Group?

Allows OCI resources to access other OCI services without user credentials.

Example: Compute instance accessing Object Storage.

26. What are OCI Policies?

Policies define permissions.

Example:

Allow group DBAdmins to manage database-family in compartment Production

27. What is OCI Vault?

Secure service for storing:

- Encryption Keys
 - Secrets
 - Certificates
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28. What is Customer Managed Key (CMK)?

Encryption keys managed by customers rather than Oracle.

Provides additional security and compliance.

29. What is Autonomous Database?

Self-managing Oracle Database that automates:

- Patching
 - Backup
 - Tuning
 - Scaling
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30. What is Oracle RAC?

Oracle Real Application Clusters allows multiple database servers to access a single database.

Benefits:

- High Availability
 - Load Balancing
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31. What is Data Guard?

Oracle disaster recovery solution.

Maintains synchronized standby databases.

32. What is OKE?

Oracle Kubernetes Engine (OKE) is OCI's managed Kubernetes service.

Benefits:

- Automated cluster management
 - High availability
 - Auto scaling
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33. What is Monitoring Service?

Tracks:

- CPU
- Memory
- Network
- Storage utilization

Provides metrics and alarms.

34. What is OCI Logging Service?

Centralized collection and analysis of logs from OCI resources.

35. What is Terraform in OCI?

Infrastructure as Code (IaC) tool used to automate OCI resource deployment.

Example:

- Create VCN
- Create Compute
- Create Load Balancer

using code.

36. What is Resource Manager?

OCI-managed Terraform service.

Benefits:

- No Terraform server required
 - Version control integration
 - Automated deployments
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37. What is OCI DevOps?

Provides CI/CD services including:

- Build Pipelines
 - Deployment Pipelines
 - Artifact Management
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38. What is OCI Cloud Guard?

Security posture management service.

Detects:

- Misconfigurations
 - Security threats
 - Compliance violations
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39. What is Security Zone?

Enforces Oracle security best practices.

Examples:

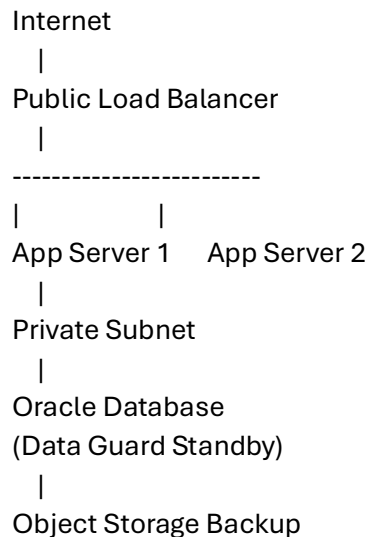
- No public database access
 - Mandatory encryption
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40. How would you design a highly available OCI architecture?

Typical design:

1. Multi-AD deployment
2. Public Load Balancer
3. Multiple Application Servers
4. Database with Data Guard
5. Object Storage backups
6. IAM and Vault security
7. Monitoring and Alarms

Architecture Flow:



Top OCI Architect Interview Questions

1. Explain OCI networking architecture.
2. Difference between Security Lists and NSGs.
3. How would you connect OCI to On-Premises?
4. Explain DRG and FastConnect.
5. Design a highly available OCI application.
6. Explain OKE architecture.
7. How do you secure OCI workloads?
8. Explain OCI IAM policies.
9. What is Cloud Guard and Security Zones?
10. How do you migrate Oracle databases to OCI?

These 40 FAQs cover approximately 80–90% of OCI Architect, OCI Administrator, and OCI Cloud Engineer interview discussions.